Ayemere J. Bellowalker

Molecular Biologist | Bioinformatician | Genomics | NGS Analysis

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UK (Open to relocation)

Professional Summary

Molecular biologist and Bioinformatician, I am a PhD candidate expected to graduate in September 2026, specialised in Translational Genomics and Multi-Omics Analysis. Experienced in PCR-based workflows, microbiome sequencing, next-generation sequencing (NGS), molecular data analysis, and reproducible bioinformatics pipelines using Python, R, Bash, and Linux. Skilled in integrating laboratory-generated and sequencing datasets to support biological discovery, assay optimisation, and translational research. Strong background in genomics, microbial systems biology, statistical analysis, and cross-functional scientific collaboration.

Core Technical Skills

Molecular Biology & Genomics
Next-generation sequencing (NGS) • PCR • qPCR • DNA extraction and quantification • Microbiome sequencing • 16S rRNA sequencing • Sample preparation • Quality control • Experimental/Assay design

Bioinformatics
Metagenomics • Transcriptomics • RNA-seq • Multi-omics integration • Differential expression analysis • Nextflow • Docker/Singularity

Computational Tools & Packages
QIIME2 • Mothur • DESeq2 • MEGAHIT • MetaBAT2 • GTDB-Tk • HISAT2 • Samtools • Bioconductor • FastQC • BLAST • SILVA database

Programming
Python • R • Bash • Linux/Unix • HPC environments • Git/GitHub • Workflow automation • Reproducible bioinformatics pipelines

Data Analysis & Statistics
Statistical modelling • Multivariate analysis • Clustering • Regression analysis • Data visualisation • Data validation & verification • Reproducible reporting

Infrastructure & Data Management
Linux • HPC • AWS EC2 • Docker • Metadata organisation • Structured data handling • Data integrity and reproducibility

Professional Experience

Doctoral Researcher - Molecular Biology, Genomics & Bioinformatics — October 2022 - September 2026

Harper Adams University - Telford, UK

- Applied biological knowledge of microbiology, genomics, host-microbe interactions, and molecular systems to guide computational analyses and experimental design.
- Designed and implemented reproducible bioinformatics workflows for the analysis of high-throughput sequencing, microbiome, transcriptomic, and multi-omics datasets.
- Analysed large-scale microbial genomics and biological datasets using R, Python, Bash, Linux, and HPC-based computational pipelines.
- Performed sequence quality control, statistical analysis, differential abundance analysis, and downstream biological interpretation of sequencing datasets.
- Processed and interpreted microbiome sequencing datasets to investigate microbial community structure, metabolism, adaptation, and host–microbe interactions.
- Developed reproducible data visualisation, reporting, and workflow documentation pipelines to support collaborative interdisciplinary research.
- Maintained organised computational records, analytical reproducibility, and structured workflow documentation across projects.
- Collaborated closely with laboratory scientists to support sequencing experiments, sample processing, molecular analyses, and interpretation of biological datasets.
- Contributed to scientific manuscripts, conference presentations, and communication of computational findings to multidisciplinary audiences.

QA Laboratory Technician - Microbiology & Pathology — July 2022 - September 2022

Intertek Food Services - Stoke-on-Trent, UK

- Processed over 200 microbiological samples daily using PCR, ELISA, and laboratory quality-control procedures within a regulated high-throughput testing environment.
- Performed sample preparation, molecular testing workflows, data recording, and result verification while maintaining compliance with GLP and ISO quality standards.
- Collaborated with multidisciplinary laboratory teams to ensure timely delivery of high-quality analytical results.

Relevant Molecular Biology Experience

- PCR, qPCR and ELISA workflows in accredited laboratory environments
- Microbiome sequencing and 16S rRNA gene analysis
- Next-generation sequencing (NGS), amplicon sequencing and RNA-seq data processing and interpretation
- RNA-seq analysis and transcriptomics workflows
- Biological sample handling and laboratory documentation
- GLP and ISO-compliant laboratory procedures
- Assay design and biological data interpretation

Selected Bioinformatics Projects

RNA-seq Differential Expression Pipeline

Bash | R | IGV | DESeq2

- Developed an end-to-end RNA-seq analysis workflow (QC → alignment → DESeq2 → interpretation).
- Processed RNA-seq data from 10 human samples across 3 conditions, achieving >85–93% alignment rates using Hisat2 and Samtools.
- Quantified expression of ~20,000+ genes, identifying 671 significant DEGs (FDR < 0.05) linked to biological conditions.
- Translated gene expression patterns into meaningful biological insights

GitHub Repository: RNA-Seq-Differential-Expression-Pipeline

Metagenomic Analysis Pipeline

Python | Nextflow | Docker | AWS | Linux | HPC

- Designed and deployed a modular pipeline processing over 2.3 billion sequencing reads across 18 samples.
- Developed scalable workflows using Nextflow, Docker, and AWS.
- Reconstructed 1,429 MAGs, including 255 medium/high-quality genomes (≥50-90% completeness).
- Identified 80 putative novel microbial genomes and predicted 469 nitrogen metabolism genes across MAGs.
- Ensured reproducibility, automation, and scalability across HPC and cloud environments.

GitHub Repository: Scalable-Metagenomic-Analysis-Pipeline

Amplicon Analysis Pipeline

Bash | R | QIIME2 | DADA2

- Analysed >5.6 million reads across 60 samples, generating 6,165 ASVs with a mean depth of ~55,000 reads/sample.
- Designed an end-to-end, reproducible pipeline integrating QIIME2 and R (phyloseq, vegan, ggplot2) for scalable microbiome data processing and visualisation.
- Identified 84 differentially abundant taxa (LDA >2.0; 11 taxa >4.0), including key microbial shifts associated with dietary starch variation, providing insights into microbiome functional responses.
- Ensured reproducibility, automation, and scalability across HPC and cloud environments.

GitHub Repository: Amplicon-Analysis-Pipeline

Education

Doctor of Philosophy (Ph.D.) October 2022 - September 2026

Harper Adams University - Telford, UK

Research Areas: Microbiome Science, Molecular Biology, Translational Genomics, Multi-Omics Data Analysis and Bioinformatics

Bachelor of Science (Honours) September 2019 - July 2022

Staffordshire University - Stoke-on-Trent, UK

1st Class Honours - Biological Science with Genomics

Conference Presentations

- First-author Poster Presentation — 14th International Gut Microbiome Symposium 2025, Clermont-Ferrand, France
- First-author Oral Presentation — MIBTP Annual Symposium 2025
- First-author Poster Presentation — MIBTP Annual Symposium 2024

Certifications & Workshops

- Linux for Genomics Workshop – Edinburgh Genomics & University of Edinburgh (2025)
- Bioinformatics and statistical analysis workshops in genomics and multi-omics data analysis

Additional Information

- Eligible for international relocation.
- Strong interest in translational genomics, molecular diagnostics and biotechnology product development.
- Strong scientific communication and technical writing skills.